

EVIDENCE-LED DECISION REPORT

# Finding Earth 2: a research-led major upgrade

How to strengthen scientific integrity, expose observation readiness, and turn the long-form article into a complete visual research workspace.

|                   |                      |                 |                  |
|-------------------|----------------------|-----------------|------------------|
| 6,354             | 16                   | 4               | 1                |
| CONFIRMED PLANETS | REPRODUCIBLE FIGURES | FOLLOW-UP LANES | OPEN METHODOLOGY |

Decision: preserve the complete NASA composite catalogue, disclose its source coherence, keep observation feasibility separate from Earth-likeness, and make every visual asset verifiably deployable under the GitHub Pages base path.

Prepared for

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Decision date: 28 August 2026  
Scope: science, provenance, follow-up utility, accessibility, performance, and static deployment

# Executive answer

The strongest upgrade is not another opaque habitability score. It is an evidence-integrity and observation-readiness release built around three connected outcomes.

## 01

### Disclose source coherence

Compare NASA's complete composite row with its coherent default published solution. Report source count, overlap, coverage, and parameter differences without treating multiple sources as an automatic penalty.

## 02

### Separate follow-up pathways

Expose transit timing, atmospheric screening, radial-velocity amplitude, and reflected-light geometry as distinct diagnostics. None changes the default Earth-2 ranking.

## 03

### Build a visual evidence workspace

Make figures base-path safe, expand the article to all reproducible visuals, use productive desktop rails, reflow on mobile, and verify the static export automatically.

## Guardrails

- Preserve a fully static GitHub Pages export; no runtime backend.
- Use public first-party archives and original papers; no paid data source.
- Never infer evidence of life, confirmed habitability, or proposal-grade feasibility.
- Keep observation feasibility out of the default Earth-likeness ranking.
- Label every assumption and preserve prior successful live-source products on catalogue-only rebuilds.

# 1 Source coherence is the highest-impact scientific gap

NASA documents two different data roles. The Planetary Systems table (ps) contains self-consistent published solution rows. The Planetary Systems Composite Parameters table (pscomppars) maximizes completeness by selecting values across references; a composite row can therefore be internally or physically inconsistent.

Decision: retain pscomppars as the catalogue spine, then disclose composite source count, whether parameters mix sources, coherent-default coverage and overlap, and the median symmetric fractional difference between overlapping values. This is context, not proof that a planet or source is wrong.

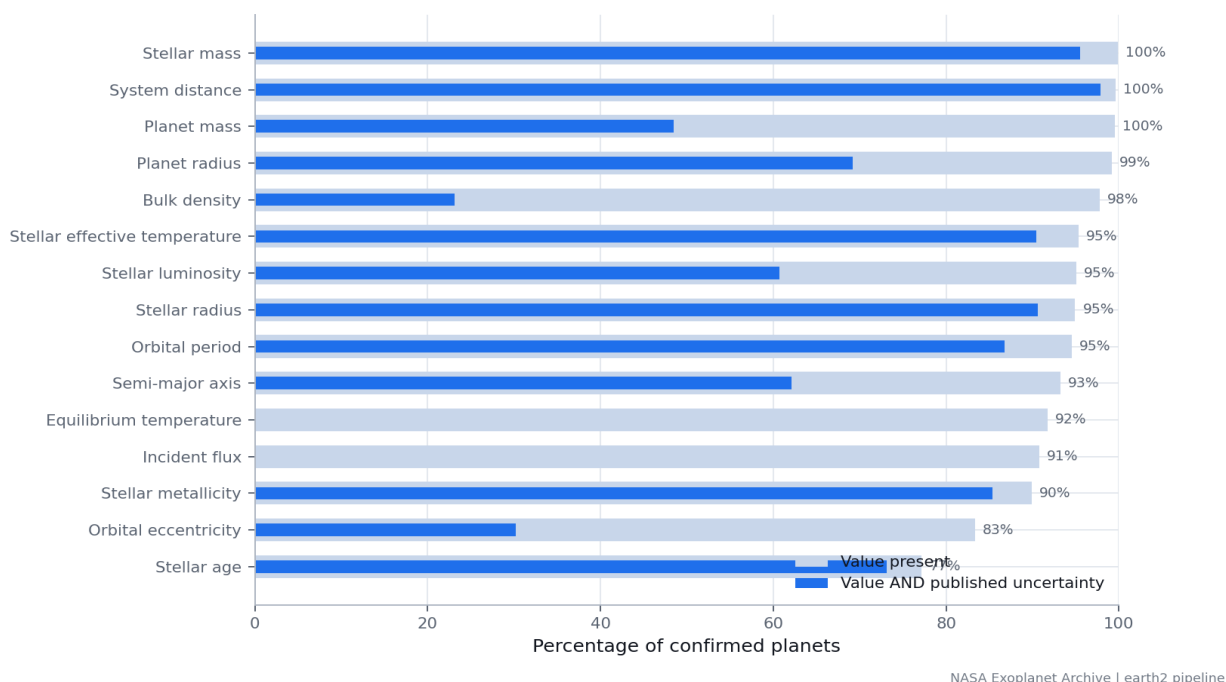


Figure 1. Archive coverage after the 28 August 2026 synchronized build. Every public product is generated from the same manifests and transformation ledger.

## Why this matters

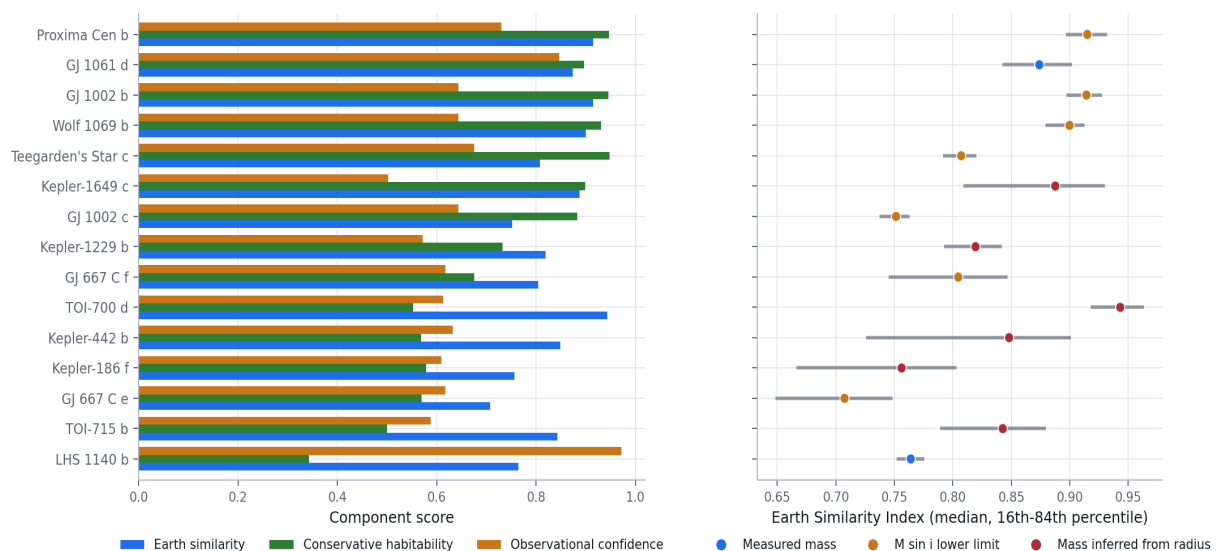
Derived properties such as equilibrium temperature, density, habitable-zone position, and Earth similarity can combine inputs published under different assumptions. The new disclosure makes that boundary visible while retaining catalogue breadth. A later research phase should propagate competing coherent solutions as a mixture, but the archive generally lacks the cross-parameter covariance required for a complete hierarchical treatment.

## 2 Observation readiness needs four separate lanes

A planet can be Earth-like in bulk properties and still be impractical for a particular observation. Conversely, a non-Earth-like planet can be exceptionally valuable for atmospheric or orbital work. Combining these into one score would hide the scientific question.

|                        |                                                                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Transit timing</b>  | Propagate period and epoch uncertainty to 2030-01-01. Flag transit timing variations and disclose that covariance is unavailable. |
| <b>Atmospheres</b>     | Show TSM and ESM as Kempton-style screening proxies, not exposure-time forecasts or guaranteed detections.                        |
| <b>Radial velocity</b> | Estimate the Keplerian semi-amplitude while separating it from stellar activity, cadence, and instrument noise.                   |
| <b>Reflected light</b> | Show maximum angular separation and an $A_g=0.30$ Lambertian quadrature contrast scenario; never label it instrument-detectable.  |

**Top Earth-2.0 candidates from the computed ranking**



NASA Exoplanet Archive | earth2 pipeline

Figure 2. The candidate ranking remains a bulk-property evidence synthesis. Follow-up lanes are displayed independently in the Observation Follow-up Lab.

### 3 The visual failure was deterministic

The article's figures existed, but root-relative image paths ignored the deployed `/finding-earth-2` base path. Next.js requires the configured base path to be reflected in local Image sources. The blank image frames were therefore a routing defect, not missing scientific output.

#### The implemented response

- Centralize public-asset URL construction and migrate images, fetch calls, and internal anchors.
- Check every exported local src and href against the production output in CI.
- Expand the article from seven visuals to the complete set of sixteen reproducible figures.
- Use a three-region desktop layout: section navigation, readable article measure, and contextual evidence rail.
- Collapse to one column on narrow screens and retain captions, text alternatives, keyboard access, and visible focus.

**Component score distributions across the analysed catalogue**

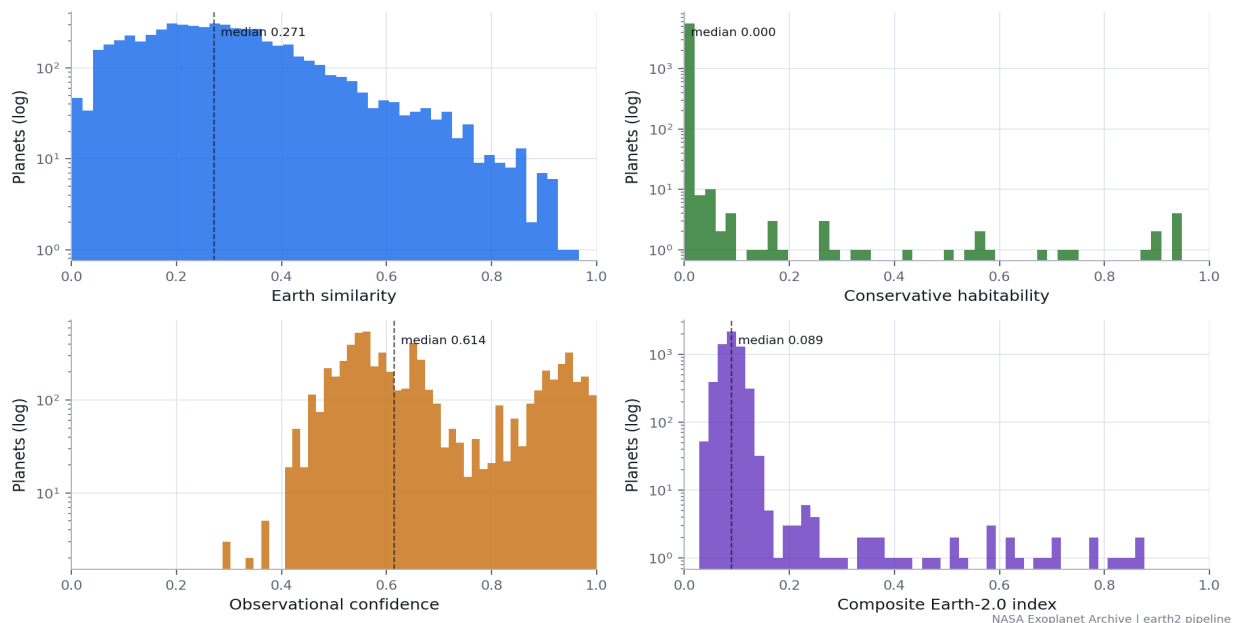


Figure 3. A titleless publication figure used by both the paper and the web narrative, preventing duplicated or clipped headings in responsive containers.

# 4 Delivered upgrade architecture

|                       |                   |                 |                                   |
|-----------------------|-------------------|-----------------|-----------------------------------|
| 6,371+                | 250               | 16              | 0                                 |
| STATIC ROUTES CHECKED | TOP ROWS PER LANE | ARTICLE FIGURES | RANKING PENALTIES FROM SOURCE MIX |

## Asset integrity

Base-path-safe images, fetches, routes, and export verification.

## Research workspace

Sticky contents, readable center measure, evidence rail, full-width stages, and mobile reflow.

## Scientific disclosure

Composite/default coherence metrics with nullable control flags and no automatic penalty.

## Observation Follow-up Lab

Searchable, lane-specific static tables for timing, atmospheres, RV, and reflected light.

## Data resilience

Catalogue-only rebuilds preserve successful archived MAST and DACE analyses with explicit provenance.

## Reproducible visuals

All sixteen figures regenerated; titleless paper variants avoid duplicate web headings.

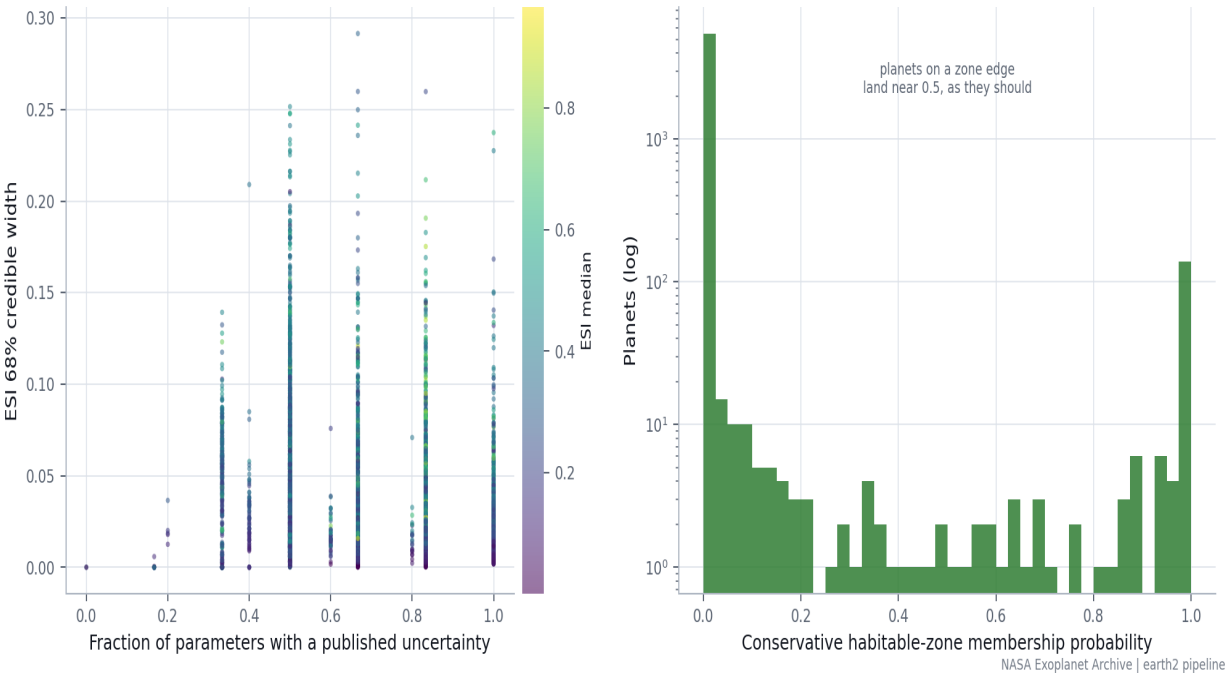


Figure 4. Uncertainty remains part of the evidence, not a decorative confidence badge. Nullable values and unavailable pathways stay explicit.

## 5 Limitations and next scientific phase

### Composite rows

Source mixing is a documented risk signal, not evidence that a row is wrong.

### Ephemerides

Linear propagation omits period-epoch covariance and cannot model unreported transit timing variations.

### TSM and ESM

Screening metrics cannot replace current ephemerides, visibility checks, saturation limits, or instrument simulators.

### Reflected light

Albedo and phase are scenario assumptions. HWO architecture remains under trade study.

### Accessibility

Automated tools catch only a subset of defects; keyboard, reflow, and scientific-description review remain manual gates.

### Recommended next phase

Model competing coherent ps solutions as an explicit mixture, quantify rank stability across solution choices, add homogeneous stellar-activity context where defensible, and connect each follow-up lane to instrument-specific simulators only when their assumptions can be recorded and reproduced.

# Claim-to-source ledger

Primary and official sources converged on the selected priorities. Links below were accessed 28 August 2026 unless a publication date is stated.

**1. NASA Exoplanet Archive.** Composite-parameter calculation notes; pscomppars may mix references while ps provides coherent published solutions.

[https://exoplanetarchive.ipac.caltech.edu/docs/pscp\\_calc.html](https://exoplanetarchive.ipac.caltech.edu/docs/pscp_calc.html)

**2. NASA Exoplanet Archive.** Planetary Systems column definitions and TAP-accessible timing, orbital, and uncertainty fields.

[https://exoplanetarchive.ipac.caltech.edu/docs/API\\_PS\\_columns.html](https://exoplanetarchive.ipac.caltech.edu/docs/API_PS_columns.html)

**3. Kempton et al. (2018).** TSM and ESM atmospheric-characterisation screening framework.

<https://arxiv.org/abs/1805.03671>

**4. Dragomir et al. (2020 revision).** TESS ephemeris maintenance and the rapid growth of timing uncertainty.

<https://arxiv.org/abs/1906.02197>

**5. Martins et al. (2013).** Reflected-light contrast dependence on radius, separation, albedo, and phase.

<https://academic.oup.com/mnras/article/436/2/1215/1125472>

**6. NASA Science.** Habitable Worlds Observatory goals and current trade-study status.

<https://science.nasa.gov/astrophysics/programs/habitable-worlds-observatory/>

**7. Next.js.** Production basePath behavior and static export constraints.

<https://nextjs.org/docs/pages/api-reference/config/next-config-js/basePath>

**8. W3C.** WCAG 2.2 normative requirements and structured alternatives for complex images.

<https://www.w3.org/TR/WCAG22/>

**9. W3C WAI.** Complex Images tutorial.

<https://www.w3.org/WAI/tutorials/images/complex/>

**10. Microsoft Playwright.** Automated accessibility testing limitations and workflow.

<https://playwright.dev/docs/accessibility-testing>

**11. Wilkinson et al. (2016).** FAIR principles for machine-readable metadata and provenance.

<https://doi.org/10.1038/sdata.2016.18>

## Research stop rationale

Further broad searching was unlikely to change the first-release scope. Remaining gaps - archive covariance, homogeneous stellar-activity data, instrument-specific noise, and final HWO parameters - are explicit research boundaries rather than inputs to speculative modeling.